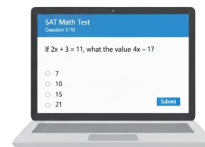
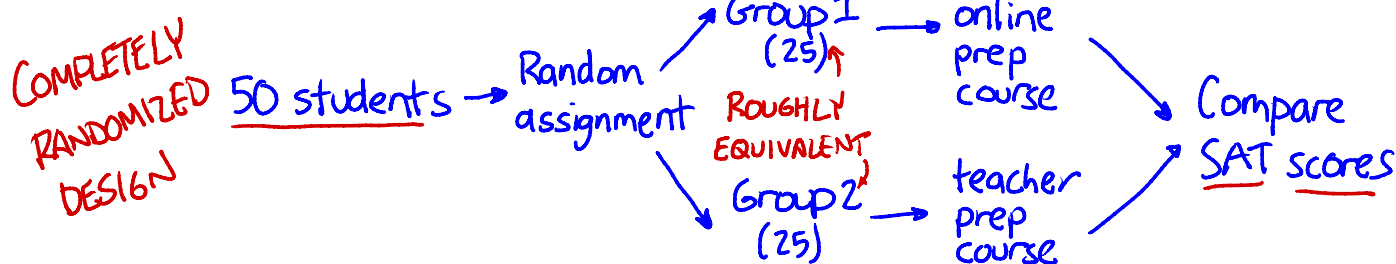


Does Type of SAT Prep Matter?



A certain high school has decided to offer an SAT prep class again this year. It will be offered in two different formats: online or classroom teacher. The counselors want to know which teaching format will yield higher SAT scores so they have allowed us to set up an experiment. 50 students have signed up to take some form of the SAT prep class.

1. Draw an outline for a completely randomized design to compare the two treatments.

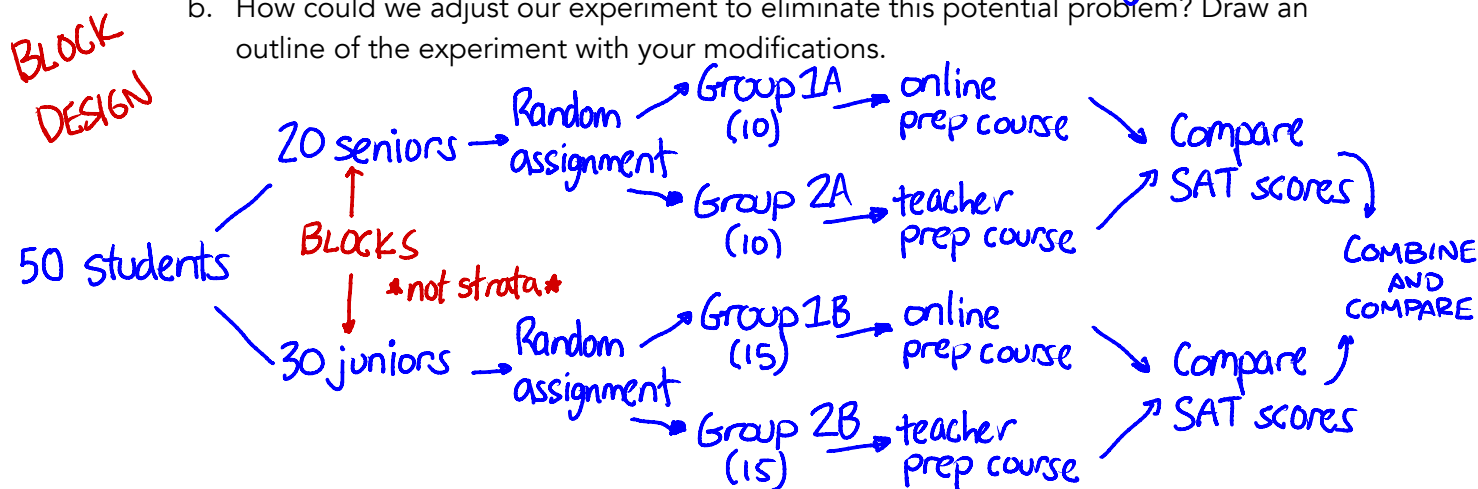


2. Of the 50 students that signed up for the prep course, 30 are juniors and 20 are seniors. The counselors are worried that the "random assignment" could potentially put all the seniors into one of the treatment groups.

- a. Why would this be a problem?

If this treatment group had higher SAT scores, we wouldn't know if it was because of the type of prep course or because it was full of seniors (seniors score higher on SAT).

- b. How could we adjust our experiment to eliminate this potential problem? Draw an outline of the experiment with your modifications.



3. The counselors are also worried that the "random assignment" could potentially put all the super hard working (high GPA) students into one of the treatment groups. They have a list of the 50 students and their GPA ranked from highest to lowest. How can they use this list to help them set up the experiment?

MATCHED PAIRS DESIGN

Pair the top two students, then the next two students, and so on (25 pairs). For each pair toss a coin. If heads, the first student in pair gets the online course. If tails → teacher course.

Lesson 2.7 – Selecting an Experimental Design

QuickNotes

LT#1 Block Design

Block - group of experimental units that are similar

Randomized Block Design

Separate subjects into blocks and then randomly assign treatments within each block.

LT#2 Matched Pairs Design

- subjects are paired and then randomly assigned to a treatment ^{block of size 2}

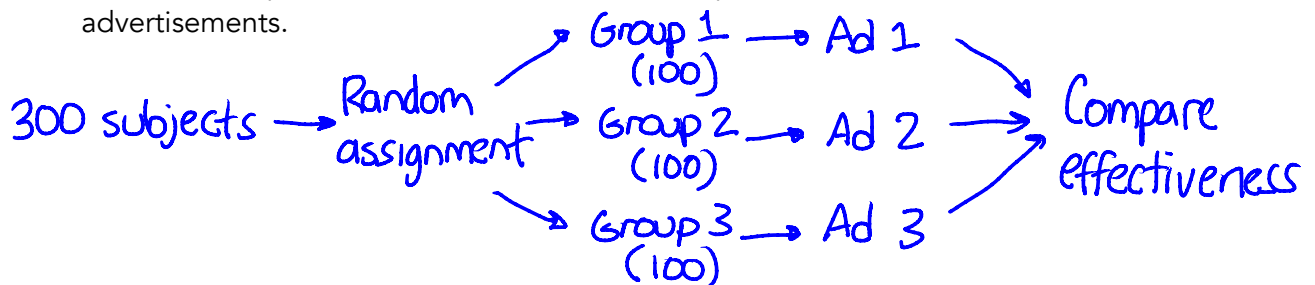
- each subject receives two treatments.

*order of the treatments must be randomized.

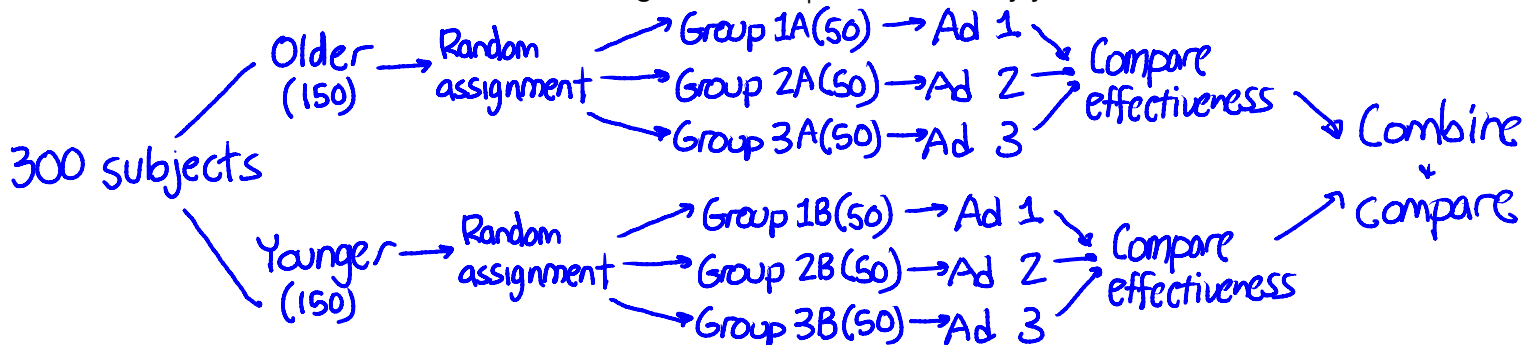
Check Your Understanding

A political strategist would like to design an experiment to compare the effectiveness of three different YouTube advertisements promoting a specific presidential candidate. She will use 300 randomly selected YouTube users for the experiment.

1. Outline a completely randomized design to compare the effectiveness of the three advertisements.



2. Outline a randomized block design for this experiment. Justify your choice of blocks.



I chose age for blocks because ads affect young people differently.

3. Why might a randomized block design be preferable in this context?

The block design helps to control age as a confounding variable. Within each block, there will be lower variability of results, making it easier to determine which ad is most effective.